

1. 讨论下列积分在指定区间上的一致收敛性:

$$(1) \int_0^{+\infty} \frac{\sin tx}{1+x^2} dx \quad (-\infty < t < +\infty);$$

$$(2) \int_0^{+\infty} e^{-t^2 x^2} \cos x dx \quad (0 < t_0 < t < +\infty);$$

$$1. (1) \left| \frac{\sin tx}{1+x^2} \right| \leq \frac{1}{1+x^2} \leq \frac{1}{x^2}, \text{ 收敛}$$

$$(2) |e^{-t^2 x^2} \cos x| \leq e^{-t_0^2 x^2}.$$

$$\text{而 } \int_0^{+\infty} e^{-t^2 x^2} dx = -1, \text{ 收敛, 故原积分一致收敛}$$

$$(3) \int_0^{+\infty} e^{-ax} \sin x dx, \quad (i) 0 < a_0 \leq a < +\infty, \quad (ii) 0 < a < +\infty \quad (\text{提示: 不论}$$

$$N \text{ 多大, 取 } A = 2k\pi > N, a = \frac{1}{A}, \text{ 有 } \left| \int_A^{+\infty} e^{-ax} \sin x dx \right| > \frac{1}{2e});$$

$$(4) \int_1^{+\infty} e^{-bx} \frac{\cos x}{\sqrt{x}} dx \quad (0 \leq b < +\infty);$$

$$(3) (i) |e^{-ax} \sin x| \leq e^{-a_0 x}, \quad \int_0^{+\infty} e^{-a_0 x} dx = -1 \quad \text{一致收敛}$$

$$(ii) \text{ 不论 } N \text{ 多大, 取 } A = 2k\pi > N, a = \frac{1}{A},$$

$$\left| \int_A^{+\infty} e^{-ax} \sin x dx \right| = \left| \int_A^{+\infty} e^{-\frac{x}{A}} \sin x dx \right| = \left| \int_1^{+\infty} A e^{-t} \sin At \cdot dt \right|$$

$$\begin{aligned} &= \left| \int_1^{+\infty} e^{-t} d \cos At \right| = \left| e^{-t} \cos At \Big|_1^{+\infty} + \int_1^{+\infty} (-\cos At) e^{-t} dt \right| \\ &= \left| e^{-1} + \int_1^{+\infty} e^{-t} d \sin At \right| \\ &= \left| \frac{1}{e} + \frac{1}{A} \left(e^{-t} \sin At \Big|_1^{+\infty} + \int_1^{+\infty} (\sin At) e^{-t} dt \right) \right| \\ &= \left| \frac{1}{e} + \frac{1}{A} \int_1^{+\infty} e^{-t} \sin At \cdot dt \right| \end{aligned}$$

则令原式 = $\varphi(t)$, 则

$$|\varphi(t)| \geq \frac{1}{e} + \frac{1}{A} |\varphi(t)| > \frac{1}{e}$$

故发散

$$(4) \quad \forall A > 1, \quad \left| \int_1^A e^{-bx} \cos x dx \right| \leq \left| \int_1^A e^{-bx} dx \right| = \left| -\frac{1}{b} e^{-bx} \Big|_1^A \right| \\ = \left| \frac{1}{b} (e^{-bA} - e^{-b}) \right| < \frac{e^{-b}}{b}$$

故 $\int_1^{+\infty} e^{-bx} \cos x dx$ 一致有界, 而 $\frac{1}{\sqrt{x}} \downarrow$ 且 $\rightarrow 0, \forall b \in [0, +\infty)$

由狄利克雷判别法, 原式收敛

$$(5) \int_0^{+\infty} t e^{-tx} dx, \quad (i) 0 < c \leq t \leq d, \quad (ii) 0 < t \leq d;$$

$$(6) \int_0^1 \frac{dx}{x^t} \quad (0 < t \leq b < 1).$$

$$(5) (i) |te^{-tx}| \leq |de^{-cx}|. \text{ 而 } \int_0^{+\infty} de^{-cx} dx = -\frac{d}{c}, \text{ 收敛}$$

此时 $\int_0^{+\infty} t e^{-tx} dx$ - 一致收敛

(ii) 对任意足够大的 N ,

$$\int_A^{+\infty} e^{-tx} dx = -e^{-tx} \Big|_A^{+\infty} = e^{-tA}$$

$$\text{取 } A = \frac{1}{t} \in [\frac{1}{d}, +\infty)$$

$$\text{则 } \int_A^{+\infty} t e^{-tx} dx = e^{-1}, \text{ 为常数}$$

则原式一定不一致收敛

$$t = \frac{1}{n}, A = 2M \\ \int_0^{+\infty} t e^{-tx} dx > \frac{1}{e^2}$$

$$(6) \left| \frac{1}{x^t} \right| \leq \frac{1}{x^b}$$

$$\text{而 } b \in (0, 1), \text{ 则 } \int_0^1 \frac{1}{x^b} dx \text{ 收敛.}$$

$$\text{故 } \int_0^1 \frac{dx}{x^t} \text{ 一致收敛}$$

2. 求下列积分的值:

$$(1) \int_0^{+\infty} \frac{e^{-ax} - e^{-bx}}{x} dx \quad (0 < a < b); \quad (2) \int_0^1 \frac{x^a - x^b}{\ln x} dx \quad (a > -1, b > -1);$$

$$(3) \int_{-\frac{1}{4}}^{+\infty} e^{-(2x^2+x+1)} dx \quad (\text{提示: 令 } t = x + \frac{1}{4});$$

$$(4) \int_0^{+\infty} \frac{e^{-x}}{\sqrt{x}} dx;$$

$$(5) \int_0^{+\infty} \frac{\sin \alpha x \cdot \cos \beta x}{x} dx \quad (\alpha > 0, \beta > 0).$$

$$2. (1) \frac{e^{-ax} - e^{-bx}}{x} = \int_b^a -e^{-xy} dy = \int_a^b e^{-xy} dy$$

$$\therefore \text{原式} = \int_0^{+\infty} dx \int_a^b e^{-xy} dy$$

$$\text{对 } f(y) = \int_0^{+\infty} e^{-xy} dx, \quad a \leq y \leq b, \quad e^{-xy} \leq e^{-ax}, \text{ 而 } \int_0^{+\infty} e^{-ax} dx \text{ 收敛}$$

则一致收敛, 可换序

$$\text{原式} = \int_a^b dy \int_0^{+\infty} e^{-xy} dx$$

$$= \int_a^b \left(-\frac{1}{y} e^{-xy} \Big|_0^{+\infty} \right) dy = \int_a^b \frac{1}{y} dy = \ln \frac{b}{a}$$

$$(2) \frac{x^a - x^b}{\ln x} = \int_b^a x^y dy, \quad \text{则原式} = \int_0^1 dx \int_b^a x^y dy$$

而 $\int_0^1 x^y dy$ 为常积分, 一致收敛.

$$\therefore \text{原式} = \int_b^a dy \int_0^1 x^y dx = \int_b^a \frac{x^{y+1}}{y+1} \Big|_0^1 dy = \int_b^a \frac{1}{y+1} dy = \ln \frac{a+1}{b+1}$$

$$(3) \int_{-\frac{1}{4}}^{+\infty} e^{-(2x^2+x+1)} dx \quad \left(\text{提示: 令 } t = x + \frac{1}{4} \right);$$

$$(4) \int_0^{+\infty} \frac{e^{-x}}{\sqrt{x}} dx;$$

$$(5) \int_0^{+\infty} \frac{\sin \alpha x \cdot \cos \beta x}{x} dx \quad (\alpha > 0, \beta > 0).$$

$$\text{令 } t = x + \frac{1}{4},$$

$$2x^2 + x + 1 = 2 \cdot \left(t - \frac{1}{4}\right)^2 + t - \frac{1}{4} + 1 = 2t^2 - t + \frac{1}{8} + t - \frac{1}{4} + 1 = 2t^2 + \frac{7}{8}$$

$$\begin{aligned} \therefore \text{原式} &= \int_0^{+\infty} e^{-(2t^2 + \frac{7}{8})} dt \\ &= e^{-\frac{7}{8}} \int_0^{+\infty} e^{-2t^2} dt \\ &= e^{-\frac{7}{8}} \cdot \frac{1}{\sqrt{2}} \int_0^{+\infty} e^{-\left(\frac{\sqrt{2}}{2}t\right)^2} d\left(\frac{\sqrt{2}}{2}t\right) \\ &= e^{-\frac{7}{8}} \cdot \frac{1}{\frac{\sqrt{2}}{2}} \cdot \frac{\sqrt{2}}{2} \\ &= \frac{\sqrt{2}}{2} e^{-\frac{7}{8}} \sqrt{2} \end{aligned}$$

$$(4) \int_0^{+\infty} \frac{e^{-x}}{\sqrt{x}} dx$$

$$\text{令 } \sqrt{x} = t, \quad x = t^2$$

$$\int_0^{+\infty} \frac{e^{-t^2}}{t} \cdot 2t \cdot dt$$

$$= 2 \cdot \int_0^{+\infty} e^{-t^2} dt$$

$$= \sqrt{2}$$

$$(5) \sin \alpha x \cos \beta x = \frac{1}{2} (\sin(\alpha + \beta)x + \sin(\alpha - \beta)x)$$

$$\frac{\sin \alpha x \cos \beta x}{x} = \frac{1}{2} \left[\frac{\sin(\alpha + \beta)x}{x} + \frac{\sin(\alpha - \beta)x}{x} \right]$$

$$\frac{1}{2} \left[\int_0^{+\infty} \frac{\sin(\alpha + \beta)x}{x} dx + \int_0^{+\infty} \frac{\sin(\alpha - \beta)x}{x} dx \right]$$

$$= \left[\frac{\pi}{2} + \frac{\pi}{2} \operatorname{sgn}(\alpha - \beta) \right] \frac{1}{2}$$

$$= \frac{\pi}{4} [1 + \operatorname{sgn}(\alpha - \beta)]$$

3. 求积分 $\int_0^{+\infty} e^{-x} \frac{\sin tx}{x} dx$ 的初等函数表达式. (提示: 先在积分号下对 t 求导.)

4. 利用 Γ 函数和 B 函数, 求下列积分的值:

$$3. f(t) = \int_0^{+\infty} e^{-x} \frac{\sin tx}{x} dx, \quad \text{易知 } \forall t \in \mathbb{R}, |e^{-x} \frac{\sin tx}{x}| \leq \frac{e^{-x}}{x}, \quad \text{而 } \int_0^{+\infty} \frac{e^{-x}}{x} dx \text{ 收敛,}$$

$\frac{1}{x} \downarrow \text{且} \rightarrow 0$

故 $\int_0^{+\infty} \frac{e^{-x}}{x} dx$ 收敛

故原式收敛.

$$\frac{\partial (e^{-x} \frac{\sin tx}{x})}{\partial t} = e^{-x} \cos tx, \quad |e^{-x} \cos tx| \leq e^{-x}, \quad \text{而 } \int_0^{+\infty} e^{-x} dx \text{ 收敛}$$

故一致收敛

又有在 $t \in \mathbb{R}$ 上连续; 故可在积分号下求导

$$\begin{aligned}
 f'(t) &= \int_0^{+\infty} e^{-x} \cdot \cos tx \cdot dx \\
 &= - \int_0^{+\infty} \cos tx \, de^{-x} \\
 &= - e^{-x} \cdot \cos tx \Big|_0^{+\infty} + \int_0^{+\infty} e^{-x} \cdot d \cos tx \\
 &= 1 - t \cdot \int_0^{+\infty} e^{-x} \cdot \sin tx \cdot dx
 \end{aligned}$$

$$\begin{aligned}
 \text{又有 } f'(t) &= \int_0^{+\infty} e^{-x} \cos tx \, dx = \frac{1}{t} \int_0^{+\infty} e^{-x} \cdot d \sin tx \\
 &= \frac{1}{t} \cdot [e^{-x} \sin tx]_0^{+\infty} + \int_0^{+\infty} \sin tx \cdot e^{-x} \cdot dx \\
 &= \frac{1}{t} \cdot \int_0^{+\infty} \sin tx \cdot e^{-x} \, dx
 \end{aligned}$$

$$\begin{aligned}
 \therefore t \cdot \int_0^{+\infty} e^{-x} \sin tx \, dx + \int_0^{+\infty} e^{-x} \cos tx \, dx \\
 = (t^2 + 1) \cdot \int_0^{+\infty} e^{-x} \cos tx \, dx = 1 \\
 \therefore f'(t) = \frac{1}{t^2 + 1}
 \end{aligned}$$

$$f(t) = \arctan t + C$$

$$\text{而 } f(0) = 0, \text{ 则 } f(t) = \arctan t$$

$$(1) \int_0^1 \sqrt[3]{x-x^2} \, dx;$$

$$(2) \int_0^{+\infty} \frac{e^{-x}}{\sqrt[3]{x}} \, dx;$$

$$(3) \int_0^a x^2 \sqrt{a^2 - x^2} \, dx \quad (a > 0); \quad (4) \int_0^{+\infty} e^{-4t} t^{3/2} \, dt;$$

$$(5) \int_0^1 \frac{dx}{\sqrt{1-x^4}} \quad (\text{提示: 令 } t = x^4); \quad (6) \int_0^{+\infty} \frac{dx}{1+x^3} \quad (\text{提示: 令 } t = \frac{1}{1+x^3});$$

$$(7) \int_0^{+\infty} \frac{x^2 \, dx}{1+x^4} \quad (\text{提示: 令 } t = \frac{1}{1+x^4});$$

$$(8) \int_0^1 \frac{x^3}{\sqrt{1-x^3}} \, dx \quad (\text{提示: 令 } t = x^3);$$

$$(9) \int_0^{\frac{\pi}{2}} \sin^m x \cos^n x \, dx \quad (m > -1, n > -1).$$

$$(1) \text{ 原式} = \int_0^1 x^{\frac{1}{3}} \sqrt{1-x} \, dx = B\left(\frac{4}{3}, \frac{4}{3}\right) = \frac{\Gamma(\frac{4}{3}) \Gamma(\frac{4}{3})}{\Gamma(\frac{8}{3})}$$

$$(2) \int_0^{+\infty} \frac{e^{-x}}{\sqrt[3]{x}} \, dx = \int_0^{+\infty} x^{-\frac{1}{3}} e^{-x} \, dx = \Gamma\left(\frac{2}{3}\right)$$

$$(3) a^2 \int_0^{\frac{\pi}{2}} \cos^2 x \cdot a^2 \sin^2 x \, dx = a^4 \int_0^{\frac{\pi}{2}} \cos^2 x \sin^2 x \, dx = \frac{a^4}{2} B\left(\frac{3}{2}, \frac{3}{2}\right) = \frac{a^4}{2} \frac{\Gamma(\frac{3}{2}) \Gamma(\frac{3}{2})}{\Gamma(3)} = \frac{1}{4} \frac{a^4}{4} \cdot 2 = \frac{1}{16} a^4 \cdot 2$$

$$(4) 4t=x, \frac{1}{32} \int_0^{+\infty} e^{-x} \cdot x^{\frac{3}{2}} \, dx = \Gamma\left(\frac{5}{2}\right) = \frac{1}{32} \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{2} = \frac{3}{128} \sqrt{2}$$

$$(5) \text{ 令 } t=x^4, \frac{1}{4} \int_0^1 t^{-\frac{3}{4}} \cdot (1-t)^{-\frac{1}{2}} \, dt = \frac{1}{4} B\left(\frac{1}{4}, \frac{1}{2}\right) = \frac{1}{4} \cdot \frac{\Gamma(\frac{1}{4}) \Gamma(\frac{1}{2})}{\Gamma(\frac{3}{4})}$$

$$\begin{aligned}
 (6) \text{ 令 } t = \frac{1}{1+x^3}, \int_1^0 t \cdot -\frac{1}{t^2} \cdot \frac{1}{3} \cdot \left(\frac{1-t}{t}\right)^{-\frac{2}{3}} \, dt &= \frac{1}{3} \int_0^1 t^{-1} \cdot \frac{(1-t)^{-\frac{2}{3}}}{t^{-\frac{2}{3}}} \, dt = \frac{1}{3} \int_0^1 t^{-\frac{1}{3}} \cdot (1-t)^{-\frac{2}{3}} \, dt \\
 &= \frac{1}{3} \cdot B\left(\frac{2}{3}, \frac{1}{3}\right)
 \end{aligned}$$

$$1+x^3 = \frac{1}{t} \quad x^3 = \frac{1}{t} - 1 \quad x = \left(\frac{1}{t} - 1\right)^{\frac{1}{3}}$$

$$(7) \text{ 令 } t = \frac{1}{1+x^4}, \quad 1+x^4 = \frac{1}{t}, \quad x = \left(\frac{1}{t} - 1\right)^{\frac{1}{4}}$$

$$\therefore \text{原式} = \int_1^0 \left(\frac{1-t}{t}\right)^{\frac{1}{2}} \cdot t \cdot \left(-\frac{1}{t^2}\right) \cdot \frac{1}{4} \left(\frac{1-t}{t}\right)^{-\frac{3}{4}} dt$$

$$= \frac{1}{4} \int_0^1 t^{-\frac{3}{4}} \cdot (1-t)^{-\frac{1}{4}} dt = \frac{1}{4} B\left(\frac{1}{4}, \frac{3}{4}\right) = \frac{1}{4} \cdot \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right) \\ = \Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{5}{4}\right)$$

$$(8) \text{ 令 } t = x^3,$$

$$\text{原式} = \int_0^1 \frac{t}{\sqrt{1-t}} \cdot \frac{1}{3} t^{-\frac{2}{3}} dt = \frac{1}{3} \int_0^1 t^{\frac{1}{3}} \cdot (1-t)^{-\frac{1}{2}} dt \\ = \frac{1}{3} \cdot B\left(\frac{4}{3}, \frac{1}{2}\right)$$

$$(9) \int_0^{\frac{\pi}{2}} \sin^m x \cos^n x dx = \frac{1}{2} \cdot B\left(\frac{n+1}{2}, \frac{m+1}{2}\right)$$